



Applying project scheduling approaches to optimize aircraft maintenance activities

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Received: 29 November 2024 / Accepted: 23 June 2026
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Abstract

We address the optimization problem of sequencing aeronautical maintenance activities, motivated by the decisions faced by a medium-sized Brazilian aeronautical maintenance center that is responsible for a fleet of dozens of aircraft. The proposed approaches accurately represent the problem and are based on integer linear programming and constraint programming. From a project scheduling perspective, they can be categorized as resource-constrained project scheduling problems, which has been previously explored in the literature in other contexts. The objective of this study is to enhance the sequencing of aircraft maintenance activities by reducing the completion time of all the activities (makespan) or the average lead time of the aircraft. Computational experiments, performed on realistic instances based on empirical corporate data, demonstrate that the proposed optimization frameworks consistently yield solutions that significantly reduce specialized labor requirements.

Keywords Aeronautical industry · Aircraft maintenance · Sequencing of activities · Integer linear programming · Constraint programming

Mathematics Subject Classification 90C90

The authors would like to acknowledge the financial support from their institutions.

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1 Introduction

The passenger transportation industry, or aviation industry, has historically been subject to significant pressure from costs. Fuel costs represent the primary concern, accounting for approximately 30% of the operating costs, followed by aircraft leasing, insurance, and maintenance costs at 20.3% (ANAC 2018). As the number of passengers required per flight to cover operating costs increases, competition for passengers drives companies to lower prices in order to achieve maximum occupancy on flights and thus guarantee their survival and some profit. It is not uncommon to observe airlines encountering financial challenges, filing for bankruptcy, or even witnessing companies posting unfavorable financial results.

In this context, the aviation industry is exerting pressure on the remainder of the supply chain in quest of efficiency, transferring some of this pressure to the manufacturers, as the efficiency of the aircraft becomes a crucial factor in the economic viability of the companies. A clear trend can be observed in the aviation industry toward increasingly efficient aircraft; one example of this is the increased use of carbon fiber composite materials, which are lighter than aluminum, in aircraft. In the most recent generations of aircraft, these materials account for up to 50% of the aircraft's weight (Boeing 2018). In addition, a new generation of engines is entering the market (geared turbo fans). These engines are designed to deliver up to 15% more fuel efficiency, with some models achieving 25% (Rolls 2018).

In the aviation industry, service centers are responsible for aircraft maintenance. Their objective is to ensure the operational readiness of their clients' fleets. Like other industries, the aviation industry is subject to considerable pressure to achieve results. Consequently, there is a growing tendency within this industry to seek solutions to these problems, with the objective of improving efficiency in terms of exploiting the knowledge and solutions that production engineering can offer. This initiative is designed to optimize the utilization of resources, whether human or equipment, in order to minimize customer downtime and reduce operating costs for the business.

Previous studies have explored various aspects of aircraft maintenance activities. Samaranyake and Kiridena (2012) proposed an integrated framework for the forward planning of aircraft maintenance activities, taking into account uncertainties. Atli and Kahraman (2012) proposed a method for minimizing the completion time of maintenance activities using a fuzzy version of the critical path model (CPM). In that work, the authors considered the CPM and replaced task execution times, which are typically deterministic or probabilistic, with a set of trapezoidal fuzzy times to address uncertainties. Junqueira et al. (2020) conducted a case study at an aircraft maintenance center in Brazil. The study reports on the use of a theory based on critical chain project management (CCPM) and CPM, with the aim of reducing the time needed to carry out preventive maintenance. Mattioda (2002) and Mattioda (2002) both applied CCPM concepts to the maintenance of large US Air Force transport aircraft.

The appropriate allocation of manpower is a problem that has been extensively explored in the literature related to aircraft maintenance. A classic example of this type of study is the work of Dijkstra et al. (1994). In this work, the authors developed a decision support system for KLM Royal Dutch Airlines. Alfares (1999) studied the allocation of maintenance labor for aircraft belonging to an oil-producing company in

Saudi Arabia. Other works on optimizing the allocation and sequencing of manpower can be seen in Butchers et al. (2001), Beliën et al. (2012), and Bertsimas et al. (2014). Ahire et al. (2000) proposed an evolution strategy algorithm for the aircraft preventive maintenance problem. The authors defined the problem as a work-force constrained preventive maintenance problem. In their study, Quan et al. (2007) examined the trade-off between manpower and makespan in preventive maintenance of complex equipment. In a maintenance center for highly complex equipment, such as aircraft, manpower is typically more expensive, making it desirable to minimize it. Safaei et al. (2011) proposed a mixed-integer programming model for sequencing maintenance activities on a fleet of military aircraft. The objective was to sequence maintenance activities on aircraft in order to maximize the number of aircraft in readiness to meet a predetermined daily flight schedule. Croteau (2015) developed a model for sequencing activities in large maintenance checks in Canada, applied to a maintenance company (Jazz Aviation).

The complexity of aeronautical maintenance scheduling often necessitates the employment of heuristic and metaheuristic approaches to overcome the computational limitations of exact methods in large-scale combinatorial optimization problems (Deng et al. 2021; Chauhan et al. 2026). Heuristic techniques, such as priority rule-based scheduling and bin packing approaches using the worst-fit decreasing (WFD) algorithm, provide efficient means to allocate maintenance tasks to available slots by treating resources as bins and tasks as items to be packed (Deng et al. 2021; Pimapunsri and Weeranant 2018). Conversely, metaheuristics such as ant colony optimization, simulated annealing, and genetic algorithms offer a more flexible exploration of the solution space, though they can struggle with local optima convergence in large instances, leading to the development of specialized hybrid algorithms involving forward and backward insertion (Eltoukhy et al. 2017). Other effective strategies include the combination of depth-first search and random search for aircraft re-assignment (Sriram and Haghani 2003) and the horizon splitting heuristic, which decomposes the planning horizon into manageable segments to improve computational performance (Kozanidis et al. 2014). Moreover, hybrid simulation–optimization remains essential for ensuring solution robustness against unstructured uncertainties in task durations (Shahmoradi-Moghadam et al. 2021). Finally, sequential heuristics are frequently utilized in line maintenance environments to minimize deviations from due dates while respecting airport resource capacities (Shaukat et al. 2020).

In this paper, we derive Integer Linear Programming (ILP) and Constraint Programming (CP) models to accurately represent the aircraft maintenance problem in a manner that reduces the completion time of the activities (makespan) or the average lead time of the aircraft, using project scheduling approaches. The approaches are based on the Resource Constrained Project Scheduling Problem (RCPSP), which has been previously explored in the literature in other contexts and initially introduced by Pritsker et al. (1969). The aim is to assist in the sequencing of maintenance activities and to analyze the effectiveness of these types of optimization approaches for solving this problem. The results of computational experiments conducted on realistic-sized examples based on real data from the case study are reported and analyzed, which indicates that the approaches are viable alternatives for practical use.

The rest of this paper is organized as follows. Section 2 describes the problem in more detail. Section 3 presents the ILP and CP models. Section 4 reports the computational results. Finally, Sect. 5 concludes with final remarks and potential paths for future research.

2 Problem description

An aircraft maintenance check, also known as a maintenance stop, is a procedure that requires grounding the aircraft to perform a series of mandatory tasks as specified in the maintenance plan. These tasks span various categories, including structure, mechanics, avionics, interiors, sealing, and painting. Depending on their scope, maintenance stops vary in duration and resource consumption. Minor checks, known as line maintenance, can be performed during short ground times, such as at the boarding gate between flights or during overnight stays at the airport. In contrast, more extensive maintenance checks, referred to as base maintenance, require the aircraft to be grounded for several days, necessitating its transport to one of the company's maintenance bases or to maintenance service provider.

The four maintenance checks illustrated in Fig. 1 are adapted from real checks carried out at a Brazilian aircraft maintenance center and are used in the computational experiments reported in Sect. 4. These have been modified solely to safeguard the company's interests while preserving the essential structure and complexity of the original operations. Each check follows a predefined precedence structure governed by technical and operational constraints. For instance, the aircraft must be electrically de-energized and hydraulic systems depressurized before avionics components are removed, and interior elements must be dismantled to provide access for structural inspections of the fuselage. Similarly, interior elements such as seats, floor panels, and sidewall linings must be taken out to provide access before structural inspections or repairs of the fuselage can be performed.

More formally, the problem of sequencing aeronautical maintenance activities can be characterized by a two-level hierarchical decision structure. At the first level, the maintenance check is divided into a set of phases, which are subject to precedence constraints that dictate their execution order. At the second level, each phase comprises a set of specific tasks. A phase is considered complete only when all its tasks have been fully executed. Each task has a fixed duration and requires a certain amount of resources during its execution. Importantly, there is a limited amount of resources available over the entire planning horizon, which are shared among all tasks being scheduled. This hierarchical arrangement extends the RCPSP, as the scheduling of individual tasks (second level) directly determines the start and completion times of the phases (first level), which in turn must respect the overall precedence network. The objective is typically to minimize the makespan, that is, to complete all phases as early as possible while satisfying both precedence and resource constraints. To exemplify, Fig. 1c illustrates the precedence network of maintenance check mc03 consisting of 9 phases and 75 tasks, with the task composition of each phase detailed in the figure. When considering the simultaneous maintenance of n aircraft, the resulting structure can be viewed as a larger precedence network formed by n such phase

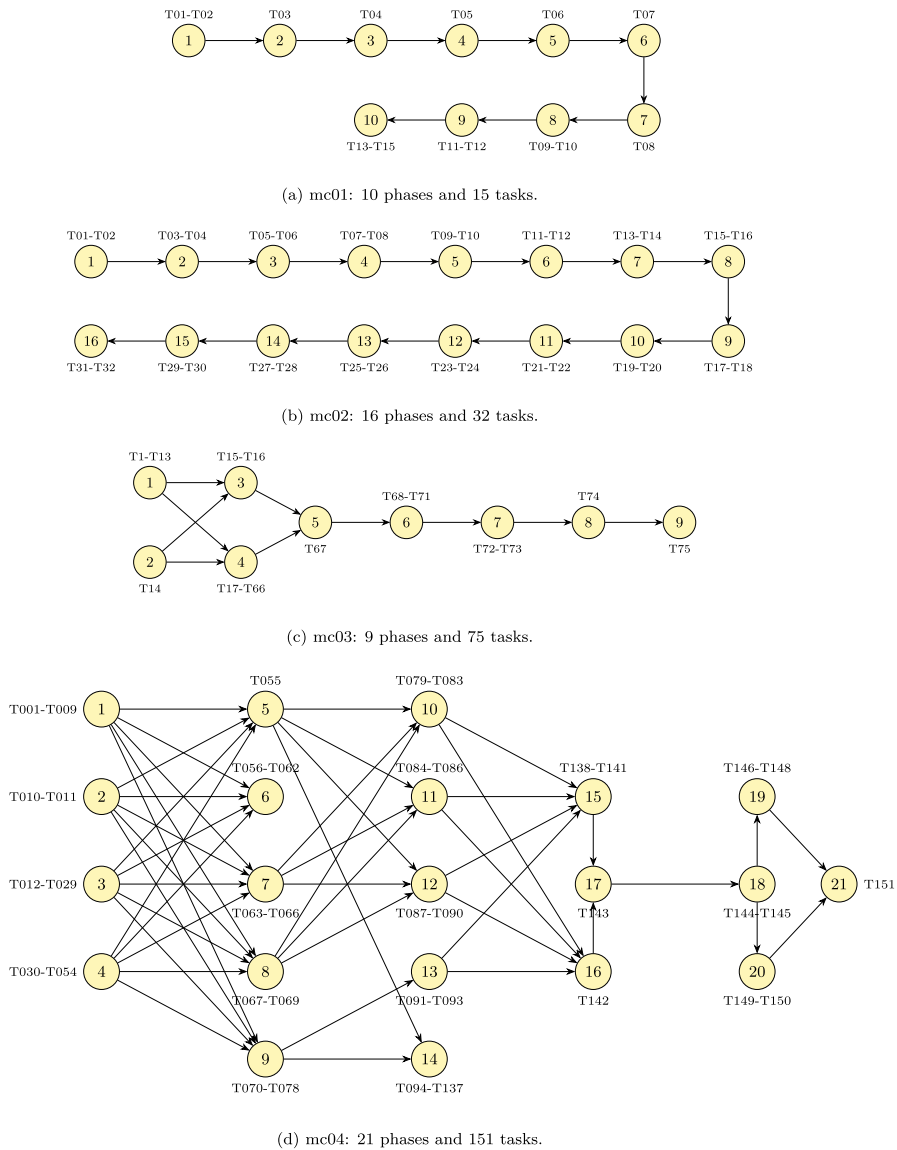


Fig. 1 Precedence networks of phases for the four maintenance checks. Each node denotes a maintenance phase, with the corresponding tasks listed alongside. Each phase is deemed complete only upon the execution of all its constituent tasks

networks running in parallel. This hierarchical decomposition enables a more granular control over resource allocation and progress tracking, facilitating the incorporation of complex operational constraints and improving the solution strategy's adaptability to real-world scenarios.

The mathematical notation used to formulate the optimization models is presented in Table 1. For the sake of clarity, concerning the planning horizon, we note that if a task starts at time instant two with a duration of three time periods, then it will end at time instant five. For the sake of feasibility, we assume that the number of workers required for any task does not exceed the number available, i.e., $n_{ik}^w \leq a^w$, for all $i \in I, k \in K_i, w \in W$.

This problem operates at the tactical or tactical-operational decision levels, focusing on determining the makespan and workforce levels required for a maintenance plan. Achieving shorter makespans to enhance aircraft availability typically demands higher labor concentration, while resource limitations may prolong maintenance duration, highlighting a critical trade-off with significant economic implications. At the tactical level, the problem considers aggregate workforce availability over the planning horizon without explicitly accounting for worker shifts or individual technician scheduling. Skill levels and experience-based efficiencies are implicitly represented through worker type classifications and task resource requirements, whereas specialized equipment availability is treated as limited shared resources. At the tactical-operational level, the problem also addresses rescheduling needs, where re-optimization with updated information can address delays and resource fluctuations to enhance robustness. While these characterizations suit the tactical scope, future extensions could incorporate shift scheduling, detailed skill matrices, and explicit equipment allocation to further improve the realism and granularity of the scheduling process.

3 Mathematical models

In this section, we present the mathematical models that accurately represent the problem of sequencing aeronautical maintenance activities. Using the mathematical notation from the previous section, we describe the ILP and CP models in Sects. 3.1 and 3.2, respectively. Additional notation for each section is limited to its scope.

3.1 An integer linear programming model

Pritsker et al. (1969) are among the first authors to propose project scheduling models with resource constraints, using a linear programming formulation with binary variables. Silva et al. (2014a, b) proposed an ILP model with the aim of minimizing the need to apply manpower in an aeronautical assembly consisting of two stages. In the second phase of this work, the authors proposed a model that considers both a labor constraint and a resource constraint. The proposed mathematical formulation, adapted from the Silva et al. (2014b)'s model, is given by model (1). This is an ILP model for a variant of the RCPSP in which time is discrete and the results are given as a function of the time t at which each task k is completed. We define the additional notation in Table 2.

$$\text{Min } z, \tag{1a}$$

Table 1 Mathematical notation of the problem

$I = \{1, 2, \dots, \bar{I}\}$	Set of aircraft in the maintenance plan
$J_i = \{1, 2, \dots, \bar{J}_i\}$	Set of phases required for the maintenance check of aircraft $i \in I$
$K_i = \{1, 2, \dots, \bar{K}_i\}$	Set of tasks required for the maintenance check of aircraft $i \in I$
$T = \{0, 1, \dots, \bar{T}\}$	Set of time instants (periods) in the discrete planning horizon
$W = \{1, 2, \dots, \bar{W}\}$	Set of worker types (resources)
$H_i \subset J_i \times J_i$	Precedence network of aircraft $i \in I$, consisting of pairs of phases $j, j' \in J_i, j < j'$, where the completion time of phase j must precede the start time of phase j'
$L_{ij} \subset K_i$	Set of tasks associated with phase $j \in J_i$ of the maintenance check of aircraft $i \in I$
$r_{ij} \in T \setminus \{\bar{T}\}$	Release time instant of phase $j \in J_i$ of aircraft $i \in I$ (the last instant is not valid as a release time)
$d_{ij} \in T \setminus \{0\}$	Due time instant of phase $j \in J_i$ of aircraft $i \in I$ (the first instant is not valid as a due time)
$p_{ik} \in \{1, 2, \dots, \bar{T}\}$	Duration of task $k \in K_i$ of aircraft $i \in I$ in time periods (minimum duration is one time period)
n_{ik}^w	Number of workers of type $w \in W$ required for task $k \in K_i$ of aircraft $i \in I$
a^w	Number of workers of type $w \in W$ available during the planning horizon

Table 2 Additional mathematical notation for the ILP model

$T_{ik} = \{t \in T \mid r_{ij} + p_{ik} \leq t \leq d_{ij}\}$	Set of feasible time instants for the completion of task $k \in L_{ij}$ of phase $j \in J_i$ of aircraft $i \in I$
s_{ij}, e_{ij}	Continuous variables for the start and completion time
$y_{ik}^t \in \{0, 1\}$	Instants of phase $j \in J_i$ of aircraft $i \in I$, respectively
$z \in \{1, \dots, \bar{T}\}$	Binary variable which equals 1 if task $k \in K_i$ of aircraft $i \in I$ finishes at time instant $t \in T_{ik}$, and equals 0 otherwise
	Integer variable representing the makespan, that is
	The completion time instant of all the phases

$$\begin{aligned}
\text{s.t. } \sum_{t \in T_{ik}} y_{ik}^t &= 1, & i \in I, k \in K_i, & (1b) \\
e_{ij} &\leq s_{ij}', & i \in I, (j, j') \in H_i, & (1c) \\
s_{ij} &\leq \sum_{t \in T_{ik}} (t - p_{ik}) \cdot y_{ik}^t, & i \in I, j \in J_i, k \in L_{ij}, & (1d) \\
e_{ij} &\geq \sum_{t \in T_{ik}} t \cdot y_{ik}^t, & i \in I, j \in J_i, k \in L_{ij}, & (1e) \\
\sum_{i \in I} \sum_{k \in K_i} \sum_{\substack{t' \in T, \\ t+1 \leq t' \leq t+p_{ik}}} n_{ik}^w \cdot y_{ik}^{t'} &\leq a^w, & w \in W, t \in T \setminus \{\bar{T}\}, & (1f) \\
z &\geq e_{ij}, & i \in I, j \in J_i, & (1g) \\
s_{ij} &\geq r_{ij}, & i \in I, j \in J_i, & (1h) \\
0 &\leq e_{ij} \leq d_{ij}, & i \in I, j \in J_i, & (1i) \\
y_{ik}^t &\in \{0, 1\}, & i \in I, k \in K_i, t \in T_{ik}, & (1j) \\
z &\in \{1, \dots, \bar{T}\}. & & (1k)
\end{aligned}$$

The objective function (1a) minimizes the project makespan across all phases of the maintenance plan. Constraints (1b) ensure that each task $k \in K_i$ of aircraft $i \in I$ is scheduled, that is, assigned a feasible completion time. Constraints (1c) represent precedence relations, ensuring that all existing precedence relationships between pairs of phases are respected. Constraints (1d) ensure that the start time of each phase $j \in J_i$ of aircraft $i \in I$ (i.e., s_{ij}) does not exceed the earliest start time among its tasks $k \in L_{ij}$. Similarly, constraints (1e) ensure that the completion time of each phase $j \in J_i$ (i.e., e_{ij}) is at least as large as the latest completion time among the corresponding tasks $k \in L_{ij}$. Constraints (1f) ensure that the availability of workers of each type $w \in W$ is respected in every time period $t \in T$ of the planning horizon. Constraints (1g) account for the makespan based on the completion times of all phases. Finally, constraints (1h) to (1k) define the domain of the decision variables.

Notice that minimizing the makespan is crucial to maximizing resource utilization and increasing the overall efficiency of the system. This is particularly relevant in environments, where the goal is to complete the work in the shortest possible total time, such as on production lines. Conversely, one could choose to minimize the average lead time of the aircraft. In this case, minimizing the average lead time is crucial to improving customer response and the efficiency of each aircraft operation. It gains relevance in environments, where the focus is on the speed of customer service or the efficiency of individual processes, such as custom manufacturing or the provision of services. While model (1) focuses on makespan minimization, model (2) presents a straightforward adaptation aimed at minimizing the average aircraft lead time. In this case, variable z is replaced by z_i , which denotes the completion time of aircraft $i \in I$, and the corresponding release time is defined as $\min_{j \in J_i} \{r_{ij}\}$.

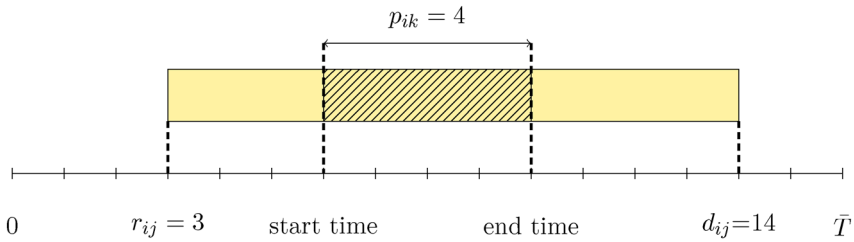


Fig. 2 Illustration of interval variable y_{ik} , $i \in I$, $j \in J_i$, $k \in L_{ij}$, with the earliest start time given by release time $r_{ij} = 3$, the latest completion time given by due time $d_{ij} = 14$, and the fixed size given by duration of $p_{ik} = 4$. A possible value for y_{ik} starts at 6 and ends at 10

$$\text{Min} \quad \frac{1}{I} \sum_{i \in I} (z_i - \min_{j \in J_i} \{r_{ij}\}), \quad (2a)$$

$$\text{s.t.} \quad (1b), (1c), (1d), (1e), (1f), (1h), (1i), (1j), \quad (2b)$$

$$z_i \geq e_{ij}, \quad i \in I, j \in J_i, \quad (2c)$$

$$z_i \in \{1, \dots, \bar{T}\}, \quad i \in I. \quad (2d)$$

3.2 A constraint programming model

As an alternative to ILP, CP is a declarative paradigm for solving combinatorial problems that are often complex and difficult to solve using traditional algorithmic approaches. It focuses on defining the problem in terms of variables, domains, and constraints. The primary goal of CP is to find a solution that satisfies all the constraints or to prove that no such solution exists. Within this framework, an interval variable is a specific type of decision variable that can take any value within a defined interval (start time, completion time, and size), as depicted in Fig. 2.

Using the OPL modeling language (Van Hentenryck 1999), we define additional notation in Table 3. The proposed mathematical formulation for the problem is given by model (3).

$$\text{Min} \quad \max_{i \in I, j \in J_i} \{\text{endOf}(x_{ij})\}, \quad (3a)$$

$$\text{s.t.} \quad \text{endBeforeStart}(x_{ij}, x_{ij'}), \quad i \in I, (j, j') \in H_i, \quad (3b)$$

$$\text{span}(x_{ij}, \{y_{ik}\}_{k \in L_{ij}}), \quad i \in I, j \in J_i, \quad (3c)$$

$$\sum_{i \in I} \sum_{k \in K_i} \text{pulse}(y_{ik}, n_{ik}^w) \leq a^w, \quad w \in W, \quad (3d)$$

$$\text{interval } x_{ij} \subseteq [r_{ij}, d_{ij}], \quad i \in I, j \in J_i, \quad (3e)$$

$$\text{interval } y_{ik} \subseteq [r_{ij}, d_{ij}], \text{ size} = p_{ik}, \quad i \in I, j \in J_i, k \in L_{ij}. \quad (3f)$$

Table 3 Additional mathematical notation for the CP model

x_{ij}	Interval variable of phase $j \in J_i$ of aircraft $i \in I$ with an earliest start value of r_{ij} And a latest end value of d_{ij}
y_{ik}	Interval variable of task $k \in L_{ij}$ of phase $j \in J_i$ of aircraft $i \in I$ with an earliest Start value of r_{ij} , a fixed duration of p_{ik} , and a latest end value of d_{ij}

The objective function (3a) minimizes the project makespan across all phases of the maintenance plan. This objective function is analogous to that of model (1). We note that predicate `endOf` returns the completion time instant of an interval variable, whereas the `max` function returns the maximum value among a list of values. Constraints (3b) are precedence constraints for ensuring that all existing precedence relations between the pairs of phases are respected. They use predicate `endBeforeStart` that ensure the first interval variable ends before the start of the second interval variable, that is, they model $\text{endOf}(x_{ij}) \leq \text{startOf}(x_{ij'})$, for each $i \in I$, $(j, j') \in H_i$. These constraints are analogous to model (1)'s constraints (1c).

The predicate `span` in constraints (3c) establishes that phase $j \in J_i$ of aircraft $i \in I$ is represented by an interval variable whose start time coincides with the earliest start among the tasks $k \in L_{ij}$ and whose completion time coincides with the latest end among these tasks. In other words, the interval associated with the phase spans all its corresponding tasks, covering them entirely in time. These constraints play a similar role as constraints (1d) and (1e) in model (1).

Similar to model (1)'s constraints (1f), constraints (3d) ensures that the quantities of each worker type $w \in W$ available are respected in any time of the planning horizon. For each worker type $w \in W$, a pulse function is defined during the execution of task $k \in K_i$ of aircraft $i \in I$ using predicate `pulse`. This function is defined from the task's start time instant to its completion time instant and consumes a certain amount of resources n_{ik}^w during this period. Furthermore, the maximum number of workers required in any time cannot exceed the value of a^w . Finally, constraints (3e) and (3f) define the domain of the decision variables. Similar to the previous section, a straightforward adaption to minimize the average lead time of the aircraft is given by model (4), where $\max_{j \in J_i} \{\text{endOf}(x_{ij})\}$ represents the completion time of aircraft $i \in I$.

$$\text{Min} \quad \frac{1}{I} \sum_{i \in I} (\max_{j \in J_i} \{\text{endOf}(x_{ij})\} - \min_{j \in J_i} \{r_{ij}\}), \quad (4a)$$

$$\text{s.t.} \quad (3b), (3c), (3d), (3e), (3f). \quad (4b)$$

It is important to note that the models presented in this section are deterministic, assuming that all parameters—such as task durations, resource availability, and precedence—are known in advance. In some real-world aircraft maintenance, however, several stochastic factors can introduce variability, including unplanned maintenance tasks discovered during inspections, delays in spare parts delivery, and fluctuations in execution times due to varying technician expertise. While not explicitly modeled here, the proposed deterministic framework provides a rigorous baseline for scheduling. In practice, these uncertainties can be partially mitigated through periodic re-optimization as new data becomes available. For more complex scenarios, this framework can be extended into stochastic or robust optimization models.

4 Computational experiments

In this section, we report the results of computational experiments carried out to evaluate the ILP models (1) and (2) of Sect. 3.1, and the CP models (3) and (4) of Sect. 3.2. These four models are denoted as ILP-Model-MS, ILP-Model-LT, CP-Model-MS, and CP-Model-LT, respectively. Note that the term -MS is used for makespan, while the term -LT is used for average lead time. All experiments were conducted on a PC with Intel Xeon E5-2680v2 (2.8 GHz), using 5 threads, 16 GB RAM, under a CentOS Linux 7.2.1511 Operating System. The models were coded in C++ using IBM CPLEX Optimization Studio v.22.1 with default settings, as a general-purpose ILP and CP solver. Each run of the solver was limited to 3600 s, unless otherwise stated.

4.1 Problem instances generated from company data

The four maintenance checks illustrated in Fig. 1 were considered for the computational experiments in this section. These instances are derived from maintenance checks ranging from 1 to 6 months at a medium-sized Brazilian aircraft maintenance center. The study considers all planned tasks within the maintenance packages; regarding unplanned tasks, those without defined working hours were excluded, while those with defined working hours were included. Data were extracted from the company's information system and adjusted by a multiplication factor to safeguard the company's interests. Subsequently, these data were converted into discrete time periods. The work packages were estimated based on the total man-hours required, and manpower efforts were calculated according to the average deadline for completing each maintenance check. Three distinct categories of manpower were required ($\bar{W} = 3$), with the number of personnel from each specialty determined for each task.

First, in Sect. 4.2, we analyze the performance of the models on four separate instances, each corresponding to an individual maintenance check (mc01, mc02, mc03, and mc04), to evaluate model scalability. Subsequently, in Sect. 4.3, we examine the models' performance when scheduling these four maintenance checks in parallel. For this aggregated instance, we assume the aircraft arrive 1 day apart, which determines the release times r_{ij} for the phases $j \in J_i$ of each aircraft $i \in I$. The due dates were set as $d_{ij} = \bar{T}$, $\forall i \in I, j \in J_i$. The experiment considered a maximum planning horizon of $\bar{T} = 690$ periods (in hours), which slightly exceeds the total time required to complete the tasks for the four aircraft if they were arranged sequentially. These data were used to generate 17 problem instances, with the available number of workers a^w varying for each worker type $w \in W$. The objective was to evaluate scenarios with both abundant and scarce labor availability to assess the resulting impact on the makespan and the average lead time of the aircraft as obtained by the models.

4.2 Results for the problem instances with a single maintenance check

We report in Table 4 the results for the single-aircraft instances when minimizing the makespan. Since these instances involve only one aircraft, the average lead time is not reported separately, as it is equivalent to the makespan in this case. The table header

includes the model (column “Model”), the problem instance (column “Instance”), the number of phases (column “Phases”), the number of tasks (column “Tasks”), and the number of time periods (column “Periods”). For each problem instance, three different scenarios for the number of available workers (a^w , $w \in \{1, 2, 3\}$) were considered. The scenario with the lowest values corresponds to the minimum workforce ensuring problem feasibility, while the highest values represent double that minimum. For each model and instance, the table presents the number of variables (column “Variables”), number of constraints (column “Constraints”), makespan (column “MS”), optimality gap (in percentage), and processing time (in seconds). In addition, the maximum and average occupancy rates (in percentage) for each worker type ($\bar{u}^w[\%]$ and $u^w[\%]$, $w \in \{1, 2, 3\}$) over the planning horizon are reported. The last row shows the average values for each column. The symbol “-” indicates that the solver with the model was unable to find a feasible solution within the time limit. In such cases, the reported average values may be not directly comparable across models.

For these experiments, the average number of variables and constraints was 11,790.25 and 723.50 for ILP-Model-MS, compared to 71.25 and 37.25 for CP-Model-MS, respectively. Since interval variables represent the entire duration in which a task can be executed, the number of variables in the CP models tends to be significantly lower than in the ILP formulations. The results in Table 4 indicate that the solver using ILP-Model-MS achieved an average optimality gap of 11.86% with a processing time of 956.12 s. In contrast, CP-Model-MS yielded a lower average gap of 2.14% and a shorter processing time of 616.62 s. Optimality was proven in 9 out of 12 instances using ILP-Model-MS, and in 10 instances using CP-Model-MS. Notably, the solver failed to find a feasible solution within the time limit for one instance when using ILP-Model-MS.

The results also serve as a sensitivity analysis with respect to workforce levels. For instance, in the most complex case (mc04 in Table 4), doubling the specialized workforce reduced the makespan from 239 to 104 periods—a 56% reduction—although some solutions have not been proven optimal. In the context of aircraft maintenance, such a reduction in turnaround time (TAT) directly translates into increased fleet availability. To illustrate the practical significance of this reduction, consider the mc04 instance, in which increasing the available workforce from the minimum feasible level to double that capacity shortened the makespan from 239 to 104 periods. Assuming each period corresponds to 1 h and that the daily opportunity cost of a mid-size business jet is USD 10,000, which lies within the conservative end of the range reported in the literature, this reduction of 135 h represents 5.625 days of additional aircraft availability. This translates into a gross revenue recovery of exactly USD 56,250. On the cost side, the workforce increase involves allocating additional technicians across the three labor categories. Taking a fully loaded hourly labor rate of USD 20 per technician, and assuming an average active increment of four technicians over the 104-h execution window, the incremental labor expenditure amounts to USD 8320. Consequently, the net benefit in this illustrative scenario is USD 47,930 per maintenance cycle. These figures are explicitly hypothetical and grounded in estimates representative of mid-size business jet MRO operations in Brazil; actual values will vary with aircraft type, contract terms, and shop configuration.

Table 4 Results for problem instances with a single maintenance check when minimizing makespan (MS) with a time limit of 3600 s

Model	Instance	Phases	Tasks	Periods	a^1	a^2	a^3	variables	constraints	MS	gap[%]	time[s]	$\bar{u}^1$ [%]	$\bar{u}^2$ [%]	$\bar{u}^3$ [%]	u^1 [%]	u^2 [%]	u^3 [%]
ILP-Model-MS	mc1	10	15	90	6	8	8	1294	334	77	0.00	0.29	83.33	87.50	75.00	15.80	22.08	21.92
					4	6	6	1294	334	82	0.00	0.28	75.00	66.67	100.00	22.26	27.64	27.44
					3	4	4	1294	334	87	0.00	0.34	100.00	100.00	100.00	27.97	39.08	38.79
	mc2	16	32	150	6	8	8	4692	577	128	0.00	1.08	83.33	87.50	75.00	15.62	23.93	25.49
					4	6	6	4692	577	142	0.00	0.93	75.00	66.67	100.00	21.13	28.76	30.63
					3	4	4	4692	577	147	0.00	1.13	100.00	100.00	100.00	27.21	41.67	44.39
	mc3	9	75	200	4	10	8	14,667	844	133	0.00	135.51	100.00	100.00	100.00	54.51	33.76	47.74
					3	7	6	14,667	844	150	0.00	83.19	100.00	100.00	100.00	64.44	42.76	56.44
					2	5	4	14,667	844	189	0.00	450.73	100.00	100.00	100.00	76.72	47.51	67.20
CP-Model-MS	mc1	10	15	90	6	8	8	26,508	1139	113	9.73	3600.00	100.00	100.00	100.00	82.96	73.10	78.76
					6	7	6	26,508	1139	160	32.57	3600.00	100.00	100.00	100.00	78.12	73.75	74.17
					4	5	4	26,508	1139	-	100.00	3600.00	-	-	-	-	-	-
	mc2	16	32	150	6	8	8	48	34	128	0.00	0.03	83.33	87.50	75.00	15.62	23.93	25.49
					4	6	6	48	34	128	0.00	0.03	83.33	87.50	75.00	15.62	23.93	25.49
					3	4	4	48	34	128	0.00	0.03	83.33	87.50	75.00	15.62	23.93	25.49
	Avg.	14.00	57.25	172.50	4.42	6.67	6.00	11790.25	723.50	128.00	11.86	956.12	92.42	91.67	95.45	44.25	41.28	46.63
					6	8	8	25	22	77	0.00	0.02	83.33	87.50	75.00	15.80	22.08	21.92
					4	6	6	25	22	82	0.00	0.03	75.00	66.67	100.00	22.26	27.64	27.44

Table 4 continued

Model	Instance	Phases	Tasks	Periods	a^1	a^2	a^3	variables	constraints	MS	gap[%]	time[s]	$\bar{u}^1$ [%]	$\bar{u}^2$ [%]	$\bar{u}^3$ [%]	u^1 [%]	u^2 [%]	u^3 [%]
mc3	9	75	200		4	6	6	48	34	142	0.00	0.13	75.00	66.67	100.00	21.13	28.76	30.63
					3	4	4	48	34	147	0.00	0.15	100.00	100.00	100.00	27.21	41.67	44.39
					4	10	8	84	22	133	0.00	0.07	100.00	100.00	100.00	54.51	33.76	47.74
					3	7	6	84	22	150	0.00	0.07	100.00	100.00	100.00	64.44	42.76	56.44
mc4	21	107	250		2	5	4	84	22	189	0.00	39.49	100.00	100.00	100.00	76.72	47.51	67.20
					8	10	8	128	71	104	0.00	159.27	100.00	100.00	100.00	90.14	79.42	85.58
					6	7	6	128	71	140	4.29	3600.00	100.00	100.00	100.00	89.29	84.29	84.76
					4	5	4	128	71	239	21.34	3600.00	100.00	100.00	100.00	78.45	69.12	74.48
Avg.		14.00	57.25	172.50	4.42	6.67	6.00	71.25	37.25	134.83	2.14	616.62	93.06	92.36	95.83	48.63	45.00	50.41

Notes: (i) parameter a^w , $w \in \{1, 2, 3\}$, denotes the number of workers of type w available over the planning horizon; (ii) MS denotes makespan; (iii) the values $\bar{u}^w$ [%] and u^w [%] represent, respectively, the maximum and average occupancy rates of worker type w (as a percentage) over the planning horizon; (iv) symbol “-” indicates that the solver was unable to find a feasible solution within the time limit, in which case the reported average values may be not directly comparable between models

To further assess the behavior of the proposed framework with respect to task duration variability, we conducted a sensitivity analysis by perturbing the task durations p_{ik} of instances mc03 and mc04 under the most resource-constrained workforce scenario. Each task duration was independently sampled from a discrete uniform distribution over three outcomes: the original duration, a 10% increase, or a 20% increase, each with equal probability of 1/3. All resulting values were rounded up to the nearest integer. To accommodate the increased workload, the planning horizon was extended by 30%, from 200 to 260 periods for mc03 and from 250 to 325 periods for mc04. CP-Model-MS was applied with a time limit of 3600 s. For mc03, the model yielded a makespan of 224 periods, with optimality proven in 8.13 s. For mc04, the model attained a makespan of 288 periods within the time limit, with an optimality gap of 21.18% after 3600 s. The perturbed instances resulted in makespan increases of 18.5% (mc03: from 189 to 224 periods) and 20.5% (mc04: from 239 to 288 periods) relative to the most resource-constrained baseline scenarios reported in Table 4. Notably, these increases are approximately twice the expected average duration perturbation of 10%, reflecting the amplifying effect of precedence and resource constraints on the overall schedule. The computational behavior remains consistent with that observed in the original instances: mc03 was solved to optimality in short processing times, while mc04 posed greater computational challenges, as expected for the most complex instance.

4.3 Results for the aggregated problem instances with four maintenance checks

We report in Tables 5 and 6 the results for the four-aircraft instances when minimizing the makespan and the average lead time of the aircraft with a time limit of 3600 s, respectively. In the header of the tables, for each instance, we present the problem instance (column “Instance”) and the number of available workers (columns a^w , $w \in \{1, 2, 3\}$). For each model and instance, we present the makespan (column “MS”), average lead time of the aircraft (column “LT”), optimality gap in percentage and processing time in seconds. We also present the average occupancy rate in percentage of each worker type (columns $u^w[\%]$, $w \in \{1, 2, 3\}$) over the planning horizon. We note that the values for makespan and average lead time have been reported for comparison purposes, despite the model in question only minimizing one of the two. The last row of each table shows the average values for each column. For the 17 problem instances, the average number of variables (resp. constraints) was 145,196 (2,894) for ILP-Model-MS, 145,199 (2,984) for ILP-Model-LT, and 285 (140) for CP-Model-MS and CP-Model-LT.

The results in Table 5 show that the solver with ILP-Model-MS has optimality gap of 32.66% and processing time of 3088.37 s, while the solver with CP-Model-MS has optimality gap of 1.56% and processing time of 2524.19 s. Optimality was proven by the solver in 4 of the 17 instances with ILP-Model-MS and in 6 instances with CP-Model-MS. As for Table 6, the results show that the solver with ILP-Model-LT has optimality gap of 30.18% and processing time of 3518.54 s, while the solver with CP-Model-LT has optimality gap of 4.63% and processing time of 3328.45 s. Optimality was proven by the solver in 2 of the 17 instances with ILP-Model-LT and in 3 instances

Table 5 Results for aggregated problem instances with four maintenance checks when minimizing makespan (MS) with a time limit of 3600s

Instance	a^1	a^2	a^3	ILP-Model-MS				CP-Model-MS				time[s]	u^1 [%]	u^2 [%]	u^3 [%]
				MS	LT	gap[%]	time[s]	MS	LT	gap[%]	time[s]				
1	20	20	20	152	115.50	0.00	541.00	152	106.75	0.00	0.21	40.56	54.47	53.16	53.16
2	19	19	19	153	117.00	0.00	706.16	153	113.75	0.00	0.43	42.41	56.97	55.59	55.59
3	18	18	18	155	118.25	0.00	1857.45	155	111.00	0.00	0.39	44.19	59.35	57.92	57.92
4	17	17	17	157	120.00	0.00	2597.61	157	119.50	0.00	50.99	46.20	62.05	60.55	60.55
5	16	16	16	162	126.00	2.47	3600.00	160	120.75	0.00	3193.01	48.16	64.69	63.12	63.12
6	15	15	15	188	111.25	17.96	3600.00	164	126.00	0.00	66.28	50.12	67.32	65.69	65.69
7	14	14	14	188	151.75	15.43	3600.00	169	132.00	1.18	3600.00	52.11	69.99	68.30	68.30
8	13	13	13	193	156.00	15.03	3600.00	176	139.00	1.14	3600.00	53.89	72.38	70.63	70.63
9	12	12	12	216	120.25	22.22	3600.00	186	147.50	1.61	3600.00	55.24	74.19	72.40	72.40
10	11	11	11	233	126.00	25.75	3600.00	196	157.00	0.51	3600.00	57.19	76.81	74.95	74.95
11	10	10	10	244	129.75	26.23	3600.00	210	171.50	0.95	3600.00	58.71	78.86	76.95	76.95
12	9	9	9	262	206.00	28.47	3600.00	226	184.25	0.44	3600.00	60.62	81.42	79.45	79.45
13	8	8	8	301	158.75	34.37	3600.00	247	202.00	0.81	3600.00	62.40	83.81	81.78	81.78
14	7	7	7	–	–	100.00	3600.00	–	199.50	2.85	3600.00	62.68	84.19	82.16	82.16
15	6	6	6	690	584.25	67.25	3600.00	324	269.25	4.63	3600.00	63.43	85.19	83.13	83.13
16	5	5	5	–	–	100.00	3600.00	–	332.25	7.93	3600.00	63.07	84.71	82.66	82.66
17	4	5	4	–	–	100.00	3600.00	–	388.75	4.41	3600.00	67.90	72.95	88.99	88.99
Avg.	12.00	12.06	12.00	235.29	167.20	32.66	3088.37	223.59	177.69	1.56	2524.19	54.64	72.31	71.61	71.61

Table 6 Results for aggregated problem instances with four maintenance checks when minimizing the average lead time of the aircraft (LT) with a time limit of 3600 s

Instance	a^1	a^2	a^3	ILP-Model-LT				CP-Model-LT									
				MS	LT	gap[%]	time[s]	u^1 [%]	u^2 [%]	u^3 [%]	MS	LT	gap[%]	time[s]	u^1 [%]	u^2 [%]	u^3 [%]
1	20	20	20	153	97.50	0.00	2646.48	40.29	54.12	52.81	153	97.50	0.00	2699.83	40.29	54.12	52.81
2	19	19	19	155	98.50	0.00	3168.69	41.87	56.23	54.87	156	98.75	0.25	3600.00	41.60	55.87	54.52
3	18	18	18	175	106.00	10.61	3600.00	39.14	52.57	51.30	158	99.50	0.00	126.33	43.35	58.23	56.82
4	17	17	17	175	104.50	6.91	3600.00	41.45	55.66	54.32	161	100.75	0.25	3600.00	45.05	60.50	59.04
5	16	16	16	172	104.50	5.65	3600.00	44.80	60.17	58.72	165	102.50	0.73	3600.00	46.70	62.73	61.21
6	15	15	15	188	109.25	8.47	3600.00	43.72	58.72	57.30	171	104.75	0.72	3600.00	48.07	64.56	63.00
7	14	14	14	197	112.00	9.26	3600.00	44.71	60.04	58.59	177	106.75	0.23	3600.00	49.76	66.83	65.21
8	13	13	13	208	115.75	10.38	3600.00	45.60	61.24	59.76	186	109.75	0.46	3600.00	50.99	68.49	66.83
9	12	12	12	216	120.25	13.61	3600.00	47.57	63.89	62.35	194	112.50	0.00	3357.47	52.96	71.13	69.42
10	11	11	11	233	126.00	19.19	3600.00	48.11	64.61	63.05	207	116.75	0.86	3600.00	54.15	72.73	70.97
11	10	10	10	244	129.75	13.62	3600.00	50.53	67.87	66.23	219	121.00	0.62	3600.00	56.30	75.62	73.79
12	9	9	9	263	139.50	17.02	3600.00	52.09	69.96	68.27	232	127.00	1.38	3600.00	59.05	79.31	77.39
13	8	8	8	301	152.50	20.17	3600.00	51.20	68.77	67.11	259	137.75	3.27	3600.00	59.51	79.92	77.99
14	7	7	7	690	590.75	78.19	3600.00	25.53	34.29	33.46	296	153.50	7.33	3600.00	59.51	79.92	77.99
15	6	6	6	–	–	100.00	3600.00	–	–	–	331	175.50	13.82	3600.00	62.08	83.38	81.37
16	5	5	5	–	–	100.00	3600.00	–	–	–	411	201.00	17.66	3600.00	60.00	80.58	78.64
17	4	5	4	–	–	100.00	3600.00	–	–	–	477	239.75	31.07	3600.00	64.62	69.43	84.70
Avg.	12.00	12.06	12.00	240.71	150.48	30.18	3518.54	44.04	59.15	57.72	232.53	129.71	4.63	3328.45	52.59	69.61	68.92

(i) parameter a^w , $w \in \{1, 2, 3\}$, denotes the number of workers of type w available over the planning horizon; (ii) MS denotes makespan and LT denotes average lead time; (iii) the values u^w [%] represent the average occupancy rate of worker type w (as a percentage); (iv) symbol “–” indicates that the solver was unable to find a feasible solution within the time limit, in which case the reported average values may be not directly comparable between models

with CP-Model-LT. Notably, the solver failed to find a feasible solution within the time limit for three instances when using ILP-Model-MS or ILP-Model-LT.

In order to assess the models' ability to provide good-quality solutions within short processing times, we report in Tables 7 and 8 the results for the four-aircraft instances when minimizing the makespan and the average lead time, respectively, using a time limit of 180 s. Overall, the number of instances without a feasible solution increased for both ILP-Model-MS and ILP-Model-LT. By contrast, the CP formulations attained notably average optimality gaps, namely, 2.30% for CP-Model-MS and 6.96% for CP-Model-LT. These findings indicate that the CP-Model-MS and CP-Model-LT can deliver good-quality solutions under tight computational budgets, a regime in which heuristic methods are often preferred; however, a direct comparison with dedicated heuristics would be necessary to draw definitive conclusions.

In general, three main behaviors can be observed in these results tables. The first is that the values for makespan and average lead time tend to increase as the values for the number of available workers a^w , $w\{1, 2, 3\}$, are reduced. The variation is not linear, given that each task consumes different quantities of worker type. Second, as expected, the average occupancy rate $u^w[\%]$, $w\{1, 2, 3\}$, tends to increase as the number of workers is reduced. These indicators are particularly pertinent for a company like the one under study, as they enable tradeoffs to be made between the amount of labor required and the project's execution time. However, this calculation is often not straightforward, beginning with the consideration of the types of workers (resources). Third, we emphasize the models based on CP had considerably better computational performance than the models based on ILP for the problem under study.

5 Conclusions

We addressed the problem of sequencing aeronautical maintenance activities motivated by the case of a medium-sized Brazilian aeronautical maintenance center. The proposed optimization approaches, based on integer programming (ILP) and constraint programming (CP) models, extend the classic Resource-Constrained Project Scheduling Problem (RCPSP) to capture the hierarchical structure and operational constraints of maintenance checks. The computational experiments demonstrated that high-quality, practically applicable solutions can be obtained with current computational resources. The models offer a viable alternative to existing planning practices by identifying feasible and improved schedules for task execution using the same workforce levels. From a computational performance perspective, the CP-based models outperformed the ILP-based models. From a practical perspective, the experiments conducted are particularly useful for a company, such as the one under study, to make tradeoffs between manpower levels and aircraft release times.

While this study focuses on model development and evaluation, we acknowledge the importance of practical deployment. The proposed optimization framework can be integrated into an aircraft maintenance scheduling software tool through: (i) a user interface connected to existing maintenance, repair and overhaul (MRO) systems for data input on tasks, resources, and precedence constraints; (ii) a real-time data update mechanism to incorporate dynamic operational information, such as resource avail-

Table 7 Results for aggregated problem instances with four maintenance checks when minimizing makespan (MS) with a time limit of 180 s

Instance	a^1	a^2	a^3	ILP-Model-MS				CP-Model-MS									
				MS	LT	gap[%]	time[s]	u^1 [%]	u^2 [%]	u^3 [%]	MS	LT	gap[%]	time[s]	u^1 [%]	u^2 [%]	u^3 [%]
1	20	20	20	166	102.50	8.43	180.00	37.14	49.88	48.67	152	106.75	0.00	0.14	40.56	54.47	53.16
2	19	19	19	171	104.50	11.11	180.00	37.95	50.97	49.74	153	113.75	0.00	0.42	42.41	56.97	55.59
3	18	18	18	175	106.00	13.14	180.00	39.14	52.57	51.30	155	111.00	0.00	0.38	44.19	59.35	57.92
4	17	17	17	177	107.25	14.12	180.00	40.98	55.03	53.71	157	119.50	0.00	48.48	46.20	62.05	60.55
5	16	16	16	182	108.50	16.48	180.00	42.34	56.87	55.49	162	125.00	1.23	180.00	47.57	63.89	62.35
6	15	15	15	-	-	100.00	180.00	-	-	-	164	126.00	0.00	66.92	50.12	67.32	65.69
7	14	14	14	197	113.50	22.84	180.00	44.71	60.04	58.59	169	132.00	1.18	180.00	52.11	69.99	68.30
8	13	13	13	208	117.00	26.83	180.00	45.60	61.24	59.76	176	139.00	1.14	180.00	53.89	72.38	70.63
9	12	12	12	216	120.25	27.36	180.00	47.57	63.89	62.35	189	150.00	3.17	180.00	54.37	73.02	71.25
10	11	11	11	-	-	100.00	180.00	-	-	-	198	159.25	1.52	180.00	56.61	76.03	74.20
11	10	10	10	244	129.75	30.68	180.00	50.53	67.87	66.23	212	173.50	1.89	180.00	58.16	78.11	76.23
12	9	9	9	263	139.50	32.59	180.00	52.09	69.96	68.27	228	168.50	1.32	180.00	60.09	80.70	78.75
13	8	8	8	-	-	100.00	180.00	-	-	-	249	185.00	1.61	180.00	61.90	83.13	81.12
14	7	7	7	-	-	100.00	180.00	-	-	-	284	203.50	3.87	180.00	62.02	83.30	81.29
15	6	6	6	-	-	100.00	180.00	-	-	-	328	266.75	5.79	180.00	62.65	84.15	82.11
16	5	5	5	-	-	100.00	180.00	-	-	-	402	348.50	10.45	180.00	61.34	82.39	80.40
17	4	5	4	-	-	100.00	180.00	-	-	-	461	392.50	5.86	180.00	66.87	71.84	87.64
Avg.	12.00	12.06	12.00	199.90	114.88	53.15	180.00	43.81	58.83	57.41	225.82	177.68	2.30	133.90	54.18	71.71	71.01

Table 8 Results for aggregated problem instances with four maintenance checks when minimizing the average lead time of the aircraft (LT) with a time limit of 180 s

Instance	a^1	a^2	a^3	ILP-Model-LT				CP-Model-LT									
				MS	LT	gap[%]	time[s]	u^1 [%]	u^2 [%]	u^3 [%]	MS	LT	gap[%]	time[s]	u^1 [%]	u^2 [%]	u^3 [%]
1	20	20	20	166	102.50	7.56	180.00	37.14	49.88	48.67	154	97.75	0.26	180.00	40.03	53.77	52.47
2	19	19	19	171	104.50	9.33	180.00	37.95	50.97	49.74	156	98.75	0.25	180.00	41.60	55.87	54.52
3	18	18	18	-	-	100.00	180.00	-	-	-	158	99.50	0.00	128.31	43.35	58.23	56.82
4	17	17	17	177	107.25	10.98	180.00	40.98	55.03	53.71	161	100.75	0.25	180.00	45.05	60.50	59.04
5	16	16	16	182	108.50	12.67	180.00	42.34	56.87	55.49	165	102.50	0.73	180.00	46.70	62.73	61.21
6	15	15	15	188	111.25	14.61	180.00	43.72	58.72	57.30	172	105.00	0.95	180.00	47.79	64.19	62.64
7	14	14	14	197	113.50	15.65	180.00	44.71	60.04	58.59	178	107.00	0.47	180.00	49.48	66.45	64.85
8	13	13	13	208	117.00	17.10	180.00	45.60	61.24	59.76	186	109.75	0.46	180.00	50.99	68.49	66.83
9	12	12	12	216	120.25	17.60	180.00	47.57	63.89	62.35	196	113.00	0.88	180.00	52.42	70.41	68.71
10	11	11	11	-	-	100.00	180.00	-	-	-	209	117.25	1.49	180.00	53.63	72.03	70.29
11	10	10	10	-	-	100.00	180.00	-	-	-	221	121.75	1.44	180.00	55.79	74.93	73.12
12	9	9	9	263	139.50	21.47	180.00	52.09	69.96	68.27	237	128.50	2.53	180.00	57.81	77.64	75.76
13	8	8	8	-	-	100.00	180.00	-	-	-	268	141.75	13.40	180.00	57.51	77.24	75.37
14	7	7	7	-	-	100.00	180.00	-	-	-	302	156.25	12.48	180.00	58.33	78.33	76.44
15	6	6	6	-	-	100.00	180.00	-	-	-	337	177.00	23.73	180.00	60.98	81.90	79.92
16	5	5	5	-	-	100.00	180.00	-	-	-	418	204.00	27.57	180.00	59.00	79.23	77.32
17	4	5	4	-	-	100.00	180.00	-	-	-	480	241.25	31.50	180.00	64.22	69.00	84.17
Avg.	12.00	12.06	12.00	196.44	113.81	54.53	180.00	43.57	58.51	57.10	235.18	130.69	6.96	176.96	52.04	68.88	68.20

(i) parameter a^w , $w \in \{1, 2, 3\}$, denotes the number of workers of type w available over the planning horizon; (ii) MS denotes makespan and LT denotes average lead time; (iii) the values u^w [%] represent the average occupancy rate of worker type w (as a percentage); (iv) symbol “-” indicates that the solver was unable to find a feasible solution within the time limit, in which case the reported average values may be not directly comparable between models

ability changes or task delays; and, (iii) a re-optimization module to generate updated schedules under changed conditions. The superior computational performance of the CP model, particularly for larger instances, makes it suitable for real-time applications, where rapid decision-making is required. The hierarchical problem structure (phases and tasks) can be directly reflected in the software interface, facilitating adoption by maintenance engineers. By enabling optimized scheduling that balances workforce allocation and task sequencing, the proposed framework can help airlines and MRO providers reduce operational costs and minimize aircraft turnaround times (TAT), thereby improving fleet availability and service reliability. Though a full software implementation is beyond this study's scope, our results demonstrate the model's viability as a decision-support tool for improving maintenance center efficiency.

Despite the promising results, some limitations of this study should be acknowledged. First, regarding heuristic comparison, no dedicated heuristic algorithm was implemented for direct benchmarking; instead, the 180-s time-limited experiments serve as a proxy for assessing solution quality under tight computational budgets. Second, concerning real-world validation, due to confidentiality constraints, the generated schedules were not validated against historical operational outcomes; thus, the reported improvements are theoretically demonstrated on realistic, anonymized data rather than confirmed through live deployment. Third, with respect to workforce dynamics, the aggregate representation of labor availability is well-suited for tactical planning but abstracts away shift patterns, skill heterogeneity, and individual technician availability that are characteristic of daily operational scheduling. Finally, regarding deployment status, a practical implementation within an operational airline or MRO setting has not yet been conducted.

For future research, several promising directions emerge. First, it is essential to validate the models against real-world historical maintenance data by comparing actual outcomes with model-generated schedules, thereby assessing their predictive accuracy and practical impact. Second, the development and evaluation of additional exact and heuristic solution methods are warranted to tackle larger-scale instances, where computational complexity becomes prohibitive. Third, and critically, extending the framework to explicitly incorporate parameter uncertainty represents a vital advancement. Future research should explore stochastic optimization and robust scheduling techniques to account for unplanned maintenance tasks (e.g., non-routine findings), delays in spare parts availability, and variability in task execution times based on technician expertise. Integrating these methods with dynamic rescheduling capabilities would enhance the resilience of aircraft maintenance planning, enabling the system to absorb operational disruptions without compromising the overall schedule integrity. Such developments would significantly improve the reliability of maintenance scheduling in complex and uncertain environments.

Acknowledgements The authors gratefully acknowledge the anonymous reviewers for their valuable comments and constructive feedback.

Funding The Article Processing Charge (APC) for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614). The authors would like to thank the financial support of the São Paulo Research Foundation (FAPESP-Brazil)

[grant number 2022/05803-3] and the National Council for Scientific and Technological Development (CNPq-Brazil) [408722/2023-1, 402492/2025-0].

Research was carried out using the computational resources of the Center for Mathematical Sciences Applied to Industry (CeMEAI) funded by FAPESP-Brazil [grant number 2013/07375-0].

Data availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest No potential conflict of interest was reported by the authors.

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